

Yalda Foroutan

PH.D. STUDENT IN COMPUTER VISION AND GRAPHICS

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RESEARCH FOCUS

Third-year Ph.D. student in the School of Computing Science at Simon Fraser University, supervised by Prof. Andrea Tagliasacchi at the GrUVi Lab. My research sits at the intersection of computer vision and computer graphics, with a focus on 3D Reconstruction, Novel View Synthesis, Robust Optimization, 360° Imagery and 3D Gaussian Splatting (3DGS).

EDUCATION

Simon Fraser University

Ph.D. in Computing Science

- Supervisor: Prof. Andrea Tagliasacchi
- Selected coursework: 3D Vision, Computer Vision

Jan 2024 – Present

Burnaby, Canada

Simon Fraser University

Ph.D. in Engineering Science

- Supervisor: Prof. Ivan V. Bajić
- Selected coursework: Information Theory, Deep Learning Systems, Stochastic Systems, Linear System Theory, Multimedia Communications, Digital Image Processing

Sep 2021 – Apr 2023

Burnaby, Canada

University of Tehran

M.Sc. in Electrical Engineering

- Thesis: Control of Computer Pointer Using Hand Gesture Recognition
- Supervisors: Prof. Ahmad Kalhor, Prof. Samad Sheikhaei
- Selected coursework: Neural Networks and Deep Learning

Sep 2017 – Sep 2020

Tehran, Iran

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Electrical Engineering (Control)

- Thesis: Analyzing Driver Consciousness Levels Using EEG Signals
- Supervisor: Prof. Mohammad A. Khosravi
- Selected coursework: Artificial Intelligence, Digital Signal Processing, Probability and Statistics, Numerical Methods, Linear Algebra, Differential Equations, Data Structures, Computer Programming, Architecture of Computers and Microprocessors, Linear Control Systems

Sep 2012 – Sep 2017

Tehran, Iran

RESEARCH EXPERIENCE

GrUVi Lab, Simon Fraser University

Research Assistant

- Researching 3D reconstruction and novel view synthesis for casual 360° captures and interactive indoor scenes over time.
- Developed a robust reconstruction model for dual-fisheye 360° imagery, addressing camera-holder visibility in casual captures.
- Designed a NeRF-guided training pipeline for 3D Gaussian Splatting that reduces dependence on SfM point-cloud initialization in complex scenes.

Jan 2024 – Present

Burnaby, Canada

Multimedia Lab, Simon Fraser University

Research Assistant

- Developed scalable image coding methods for humans and machines, focusing on base-layer and enhancement-layer efficiency.
- Contributed to learnable compression and compressed-vision projects that led to ICIP, ICMEW, and TPAMI publications.

Feb 2022 – Apr 2023

Burnaby, Canada

Advanced Circuits for Data Communication Lab, University of Tehran

Research Assistant

- Built a hand-gesture recognition application for computer-pointer control and collected a four-gesture hand dataset.
- Worked on object detection and image classification components for gesture-based human-computer interaction.

Feb 2018 – Sep 2020

Tehran, Iran

PUBLICATIONS

FullCircle: Effortless 3D Reconstruction from Casual 360° Captures

Yalda Foroutan, Ipek Oztas, Daniel Rebain, Aysegul Dundar, Kwang Moo Yi, Lily Goli, Andrea Tagliasacchi

arXiv, 2026 [📄](#) paper [🔗](#) project [</>](#) code

- Built a reconstruction pipeline for casual dual-fisheye 360° images, targeting everyday captures rather than controlled capture rigs.
- Focused on robust optimization under challenging visibility, including the camera-holder artifact common in handheld 360° footage.

Evaluating Alternatives to SfM Point Cloud Initialization for Gaussian Splatting

Yalda Foroutan, Daniel Rebain, Kwang Moo Yi, Andrea Tagliasacchi

arXiv, 2024 [📄](#) paper

- Evaluated alternatives to SfM point-cloud initialization for Gaussian Splatting and used NeRF guidance to improve optimization.
- Studied reconstruction behavior in scenes with sparse views and dynamic or difficult-to-match content.

Rate-Distortion Theory in Coding for Machines and its Application

Alon Harell, Yalda Foroutan, Nilesch Ahuja, Parual Datta, Bhavya Kanzariya, V. Srinivasa Somayazulu, Omesh Tickoo, Anderson de Andrade, Ivan V. Bajić

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025 [📄](#) paper

Base Layer Efficiency in Scalable Human-Machine Coding

Yalda Foroutan, Alon Harell, Anderson de Andrade, Ivan V. Bajić

IEEE International Conference on Image Processing (ICIP), 2023 [📄](#) paper

VVC+M: Plug and Play Scalable Image Coding for Humans and Machines

Alon Harell, Yalda Foroutan, Ivan V. Bajić

IEEE International Conference on Multimedia and Expo Workshops (ICMEW), 2023 [📄](#) paper

Conditional and Residual Methods in Scalable Coding for Humans and Machines

Anderson de Andrade, Alon Harell, Yalda Foroutan, Ivan V. Bajić

IEEE International Conference on Multimedia and Expo Workshops (ICMEW), 2023 [📄](#) paper

Control of Computer Pointer Using Hand Gesture Recognition in Motion Pictures

Yalda Foroutan, Ahmad Kalhor, Saeid Mohammadi Nejati, Samad Sheikhaei

arXiv, 2020 [📄](#) paper

SELECTED ACADEMIC PROJECTS

3D Vision

2024

- Moving Least Squares surface reconstruction, 3D Morphable Model face fitting;
- Rigid ICP, non-rigid fitting, and neural field reconstruction.
- Practiced geometry-processing and optimization workflows for point clouds, meshes, and implicit neural representations.

Simon Fraser University

Computer Vision

2023

- Digit recognition with convolutional neural networks; image classification with PyTorch.
- Object detection, semantic segmentation, and instance segmentation.
- Planar homographies and 3D reconstruction.

Simon Fraser University

Deep Learning

2020 – 2023

- Finding the minimum bitrates for a computer vision task.
- Designed a CNN classifier for Fashion-MNIST and implemented an RNN for stock market prediction.
- Built RNN-based Shakespeare text generation.
- Implemented DCGAN, WGAN, and ACGAN on Fashion-MNIST and CIFAR-10.

Simon Fraser University / University of Tehran

Other Python Projects

2018 – 2023

- CANF-VC limitations for shot transitions, motion information, and reference frames.
- Breast cancer image segmentation.
- Face filters for eye color and size; 3D sunglasses overlay.
- Low-light enhancement, super-resolution, and related image-quality experiments with GANs.

Simon Fraser University / University of Tehran

SKILLS

Programming	Python, MATLAB, C/C++, Julia, Bash, R
Frameworks and Libraries	PyTorch, TensorFlow, OpenCV
Tools	Git, HTML/CSS

TEACHING EXPERIENCE

2026	Computer Vision	Simon Fraser University
2025	Computer Vision	Simon Fraser University
2024	Introduction to Engineering Analysis	Simon Fraser University
2023	Capstone B	Simon Fraser University
2023	Graphical Communication	Simon Fraser University
2023	Capstone A	Simon Fraser University
2023	Computational Neuroscience (Certificate)	Neuromatch Academy
2022	Graphical Communication	Simon Fraser University
2021	Computational Neuroscience (Certificate)	Neuromatch Academy
2021	Deep Learning (Certificate)	Neuromatch Academy
2020	Neural Networks and Deep Learning	University of Tehran
2020	Electronics I	University of Tehran
2019	Electronics I	University of Tehran
2019	Electronics III	University of Tehran

HONORS AND AWARDS

2026	Computing Science Graduate Fellowship	School of Computing Science, Simon Fraser University, Canada
2026	Graduate Fellowship	Simon Fraser University, Canada
2026	Representation learning & GenAI Winter School	Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE
2025	Computing Science Graduate Fellowship	School of Computing Science, Simon Fraser University, Canada
2025	Graduate Fellowship	Simon Fraser University, Canada
2025	Ph.D. Research Scholarship	Simon Fraser University, Canada
2024	Computing Science Graduate Fellowship	School of Computing Science, Simon Fraser University, Canada
2024	Graduate Fellowship	Simon Fraser University, Canada
2024	Ph.D. Research Scholarship	Simon Fraser University, Canada
2023	Upper Bound Talent Bursary	Alberta Machine Intelligence Institute (Amii), Canada
2020	Direct Ph.D. Admission	University of Tehran, Iran
2008	2nd Place in Iran National Mathematics Olympiad	PAYA Mathematics Olympiad, Iran
2008	Exceptional Talent Recognition (High School)	National Organization for Development of Exceptional Talents, Iran
2006	1st Place in Regional Mathematics Olympiad	Abadan Mathematics Olympiad, Iran
2005	Exceptional Talent Recognition (Middle School)	National Organization for Development of Exceptional Talents, Iran